Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction

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UNDERTAKING

We declare that the work presented in this report titled "*Neuro-Symbolic Reasoning over Patient-Specific Knowledge Graphs for Breast Cancer Progression Prediction*", submitted to the Computer Science and Engineering Department, Motilal Nehru National Institute of Technology Allahabad, Prayagraj, for the award of the ***Bachelor of Technology*** degree in ***Computer Science & Engineering***, is our original work. We have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, we accept that our degree may be unconditionally withdrawn.

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CERTIFICATE

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Abstract

Predicting whether and when a breast cancer will progress after treatment is a central question in oncology, and a prediction is only useful to a clinician if it can be explained. This project builds patient-specific knowledge graphs for the 1,083 breast invasive carcinoma patients of the TCGA PanCancer Atlas 2018 cohort and fuses each one with PrimeKG, a biomedical knowledge graph of 8.1 million relations. Every patient graph links the patient to their tumour sample and to their mutated and copy-number-altered genes from a 500-gene panel. These genes are joined to PrimeKG pathways, drugs, and breast-disease entities through their NCBI Entrez identifiers.

On these fused graphs we benchmark seven graph neural network architectures (GCN, GraphSAGE, GATv2, R-GCN, RGAT, GIN, and a graph transformer) under three objectives (binary cross-entropy, a differentiable concordance index, and the Cox partial likelihood). We also add patient-conditioned graph structure learning. Discrimination saturates near a concordance index of 0.67 across every model family, and the clinical variables carry most of the signal.

We therefore turn the system into a neuro-symbolic one. A survival GNN supplies neural edge importances. For the symbolic part we go beyond IF–THEN rule induction and compare four survival-aware reasoners over 422 first-order ground atoms, including relational PrimeKG atoms such as $\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, w)$: a Markov logic network whose formula weights are learned with the Cox partial likelihood, probabilistic soft logic with a collective patient-similarity rule, a policy-gradient rule generator with a replication reward, and a hybrid in which reinforcement learning proposes the Markov logic formulas. The hybrid is the rule engine. A fused Cox model, chosen by the one-standard-error rule, produces each patient's risk

and survival curve. Accelerated failure time models fitted on progressors estimate the month of progression. Each explanation path is scored as 0.6 times its neural importance plus 0.4 times the weights of the learned formulas it grounds, and a prominence cutoff selects the top-K paths.

The system writes a hospital-style PDF report for every patient, grouped by recorded progression timeline. All predictions are out-of-fold. The Markov logic engines reach a concordance of 0.63 against 0.62 for RuleKit, with 69% of their formulas reasoning over PrimeKG relations against 10% of RuleKit's rules. The fused model matches the best clinical model (concordance 0.664 against 0.675, within one standard error). The progression-month estimate is within 12 months of the recorded month for 48% of progressors, against 37% for a constant baseline. Every report grounds its risk estimate in explicit knowledge-graph paths and weighted logic formulas.

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We express our sincere gratitude to our supervisor, Dr. Shashwati Banerjea, for guidance, patience with our many iterations, and above all for steering the project from raw model accuracy towards neuro-symbolic reasoning and interpretable patient reports. This feedback shaped every milestone of this work.

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Contents

Abstract **iv**

Acknowledgements **vi**

1 Introduction **1**

1.1 Motivation 2
1.2 Problem Statement 2
1.3 Objectives 3
1.4 Contributions 3
1.5 Organisation of the Report 4

2 Related Work **5**

2.1 Cancer Genomics Data 5
2.2 Biomedical Knowledge Graphs 5
2.3 Graph Neural Networks 6
2.4 Survival Analysis 6
2.5 Rule Learning for Survival Data 7
2.6 Statistical Relational Learning: Markov Logic and Soft Logic 7
2.7 Reinforcement Learning for Rules and Paths 8
2.8 Neuro-Symbolic AI 8
2.9 Gene-Expression Prognosis Cohorts 8

3 Proposed Work **9**

3.1 Data 10

3.2 Patient Graph Construction 10
3.3 Gene Panel 11
3.4 Fusion with PrimeKG 12
3.5 Progression Labels 12
3.6 Graph Neural Network Models 13
3.6.1 Training Objectives 14
3.7 Patient-Conditioned Graph Structure Learning 14
3.8 Neuro-Symbolic Progression Model 15
3.8.1 Neural Component: Survival GNN 15
3.8.2 Symbolic Baseline: Predicates and RuleKit Rules 15
3.8.3 Knowledge-Graph Fact Evidence 16
3.8.4 Logic Reasoners: Beyond IF–THEN Rules 16
3.8.5 Fusion and Model Selection 18
3.8.6 Estimating the Month of Progression 19
3.8.7 Neuro-Symbolic Path Scoring and Top-K Selection 19
3.9 Report Groups 20
3.10 Patient Report Design 20

4 Experimental Setup and Results Analysis 22
4.1 Experimental Setup 22
4.2 Milestone 1: Patient Graphs 23
4.3 Milestone 2: PrimeKG Fusion and the 36-Month Benchmark 23
4.4 Concordance Training and the Architecture Zoo 25
4.5 Graph Structure Learning 25
4.6 External Benchmark on GSE2034 26
4.7 Neuro-Symbolic Model 26
4.7.1 Cohort 26
4.7.2 Discrimination and Model Selection 27
4.7.3 Symbolic Reasoners Compared 29
4.7.4 Estimating the Month of Progression 31
4.7.5 Neuro-Symbolic Paths 33
4.7.6 Patient Reports 33

4.8 Discussion 34

5 Conclusion and Future Work **41**

5.1 Conclusion 41

5.2 Limitations 42

5.3 Future Work 42

A Reproducibility **44**

References **46**

List of Tables

1 Report groups by recorded progression-free survival. 20
2 36-month progression, five-fold mean ± standard deviation. 23
3 36-month endpoint: AUROC, AUPRC, and Harrell's C (five-fold means) under two objectives. 25
4 Full PFS follow-up (1,083 patients, 145 events). Harrell's C is the five-fold mean ± standard deviation. Uno's C is truncated at 120 months, and IBS is the integrated Brier score to 120 months (lower is better). 29
5 Symbolic reasoners on the full follow-up: five-fold Harrell's C (mean ± sd), Uno's C, learned formulas per fold, and the share of formulas that reason over a PrimeKG relation. 29
6 Estimating the month of progression on held-out progressors. 32
7 Main source modules of the neuro-symbolic system. 45

List of Figures

1 Overview of the neuro-symbolic progression system. 9
2 A Milestone 1 patient graph (TCGA-A2-A0SW): one patient, one tumour sample, and its mutated genes. 11
3 Part of a fused graph: altered genes of one patient connected to PrimeKG partner proteins, pathways, and a breast-disease node. . . . 13
4 Out-of-fold ROC curves of the graph models on the 36-month endpoint. 24
5 Clinical AUROC as a function of the progression window. 24
6 Graph structure learning against the fixed two-hop graph on the 36-month endpoint. 26
7 Recorded progression months, with the report group boundaries at 12, 24, and 36 months. 27
8 Fold-mean Harrell's C-index of every model. The report model is highlighted. 28
9 Fold-mean Harrell's C of the symbolic reasoners. The engine is highlighted. 30
10 Kaplan–Meier curve of the patients for whom the RL-proposed formula $\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{VEGFA-VEGFR2}) \wedge \mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{Ion homeostasis})$ holds, against the cohort. 31
11 Model-estimated against recorded progression month. Bars are the model's interquartile ranges, and the diagonal is perfect agreement. . 32
12 Distribution of neuro-symbolic path scores, with the prominence cut-off at the cohort's 80th percentile. 33

13 Top-K paths of patient TCGA-BH-A1EV. The bar on the right splits each NS score into its neural part (navy) and its rule part (red: the grounded formulas raise the hazard; blue: they lower it). 34
14 Patient report, page 1: record, progression assessment, survival curve, and summary. 35
15 Patient report, page 2: top-K knowledge-graph paths and the weighted logic formulas that fire. 36
16 Patient report, page 3: molecular profile, PrimeKG drug links, similar patients, and reading notes. 37
17 First page of a report in group N (no progression recorded). 38
18 Summary page of group A, progression within 12 months. 39

Chapter 1

Introduction

Breast cancer is the most commonly diagnosed cancer worldwide, with an estimated 2.3 million new cases in 2020 [1]. Its course after primary treatment varies widely. Some patients never relapse, some progress within months, and some progress a decade later. Clinicians estimate this risk from pathological stage, lymph-node status, and molecular subtype [2]. Genome-scale profiling, however, records hundreds of somatic alterations per tumour, and most of them have no established prognostic meaning on their own.

Biomedical knowledge graphs offer a way to read those alterations in context. PrimeKG [3] integrates twenty curated resources into a single graph of 129,375 nodes and over 8 million relations. It connects genes and proteins to pathways, drugs, diseases, and phenotypes. An altered gene in a tumour can therefore be connected to the pathways it participates in, to the drugs that target it, and to the breast diseases it has been linked to. Graph neural networks (GNNs) [4, 5] can learn from such structure, and explainers such as GNNExplainer [6] can say which parts of a graph drove a prediction.

1.1 Motivation

A prognostic model is only useful at the bedside if a clinician can see why it reached its conclusion. Purely neural models over large graphs give a score without reasons. Purely symbolic models, such as survival rules, are readable but cannot weigh thousands of interacting facts. Neuro-symbolic AI [7] combines the two. Neural components learn from data, and symbolic components state knowledge explicitly and can be checked.

This project started as a pure GNN benchmark. After several iterations, including relation-aware architectures, concordance training, and graph structure learning, its predictive performance saturated. Following our supervisor's direction, the final phase builds a neuro-symbolic layer and turns its output into per-patient reports. The reports explain every prediction with knowledge-graph paths and weighted logic formulas, and they group patients by when they progressed.

1.2 Problem Statement

Given the clinical record and the somatic mutation and copy-number profile of a breast cancer patient, together with the PrimeKG biomedical knowledge graph, the system must:

1. estimate the patient's risk of disease progression and their progression-free survival curve;
2. estimate the month in which progression occurs, given that it occurs;
3. explain the estimate through explicit, verifiable knowledge-graph paths and weighted logic formulas; and
4. present the result as a readable clinical report, grouped by progression timeline.

Outcome variables must never leak into the model inputs. Every reported prediction must come from a model that did not see that patient during training.

1.3 Objectives

- Build validated patient-specific graphs from the cBioPortal TCGA BRCA Pan-Cancer Atlas 2018 study [8, 9].
- Fuse each patient graph with a filtered PrimeKG neighbourhood through NCBI Entrez identifiers.
- Benchmark GNN architectures and training objectives on leakage-safe progression labels.
- Build a neuro-symbolic model that combines a survival GNN with survival-aware logical reasoning, comparing Markov logic [38], probabilistic soft logic [40], and reinforcement-learned rules against a RuleKit baseline [10].
- Estimate the month of progression and evaluate it against the recorded month.
- Generate a hospital-style PDF report for every patient, binned by progression timeline.

1.4 Contributions

1. A reproducible pipeline from raw cBioPortal tables to validated, leakage-checked patient graphs and PrimeKG-fused PyTorch Geometric graphs, with 58 unit tests.
2. A benchmark of seven GNN architectures under three objectives, a horizon sweep, graph structure learning, and an external microarray benchmark, all with out-of-fold evaluation.
3. A neuro-symbolic survival model. Its symbolic engine is a *survival Markov logic network* whose first-order formulas, including multi-hop PrimeKG conjunctions, are proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, and whose weights are learned with the Cox partial likelihood. It is compared with survival probabilistic soft logic

with collective inference, RL rules alone, and RuleKit. Explanation paths are scored as 0.6 times their GNNExplainer importance plus 0.4 times the weights of the formulas they ground.

4. An honest model-selection procedure (the one-standard-error rule) and a progression-month estimator evaluated against recorded outcomes.

5. 1,083 patient reports, group summaries, a cohort overview, and an executed notebook that documents the work.

1.5 Organisation of the Report

Chapter 2 reviews related work on cancer genomics data, knowledge graphs, graph neural networks, survival analysis, and rule learning. Chapter 3 describes the proposed system, from data to reports. Chapter 4 gives the experimental setup and analyses the results of every milestone. Chapter 5 concludes and outlines future work. The appendix lists the commands that reproduce every result.

Chapter 2

Related Work

2.1 Cancer Genomics Data

The Cancer Genome Atlas (TCGA) PanCancer Atlas re-analysed more than 10,000 tumours from 33 cancer types with uniform pipelines [11]. Liu et al. [12] curated its clinical data into a resource with four standardised endpoints: overall survival, disease-specific survival, disease-free interval, and progression-free interval. They recommended progression-free interval for breast cancer because of its event count and follow-up. cBioPortal [8, 9] distributes these data as tab-delimited study bundles. Copy-number calls in these bundles are discretised by GISTIC2.0 [13] into homozygous deletion, hemizygous deletion, neutral, gain, and high-level amplification. The PAM50 intrinsic subtypes [2] (Luminal A, Luminal B, HER2-enriched, Basal-like, Normal-like) are the standard molecular classification of breast cancer.

2.2 Biomedical Knowledge Graphs

PrimeKG [3] is a precision-medicine knowledge graph built from 20 resources, including Reactome pathways [14], DrugBank drug targets [15], and curated disease-gene associations. It describes 17,080 diseases with 4,050,249 relations. Its nodes keep source identifiers, such as NCBI Entrez for genes, so patient data can be aligned to

it without fuzzy name matching. Knowledge graphs supply the symbolic, human-readable layer that neuro-symbolic systems reason over.

2.3 Graph Neural Networks

Graph convolutional networks (GCN) [4] propagate degree-normalised neighbour features. GraphSAGE [5] learns separate self and neighbour transforms and generalises inductively to unseen graphs, which suits a setting with one graph per patient. Graph attention networks learn a weight per edge, and GATv2 [16] fixes the static-attention limitation of the original formulation. Relational GCN (R-GCN) [17] and relational graph attention (RGAT) [18] give each relation type its own weights, which is the classical treatment of heterogeneous knowledge graphs. The graph isomorphism network (GIN) [19] uses sum aggregation and is as expressive as the Weisfeiler–Lehman test. TransformerConv [20] brings multi-head dot-product attention with edge features to message passing. All of these are implemented in PyTorch Geometric [21]. GNNExplainer [6] learns a soft mask over edges that preserves a model's prediction, which yields an importance score per edge.

Graph structure learning [22, 23] learns which edges a GNN should use, instead of taking the input graph as given. This is relevant when a knowledge-graph neighbourhood may be too large or too noisy.

2.4 Survival Analysis

Progression is a time-to-event outcome with right censoring: a patient without a recorded progression may still progress after their last follow-up. The Kaplan–Meier estimator [24] estimates a survival curve non-parametrically. The log-rank test [25] compares the curves of two groups. The Cox proportional hazards model [26] relates covariates to a hazard ratio and is fitted by maximising the partial likelihood. Accelerated failure time (AFT) models [27] instead model the logarithm of the event time directly. Random survival forests [28] extend random forests to censored data. DeepSurv [29] trains a neural network on the Cox partial likelihood,

the approach our survival GNN follows.

Discrimination is measured by Harrell's concordance index [30]: among comparable pairs of patients, the fraction in which the one who progressed earlier received the higher risk. Uno's C [31] reweights pairs by the inverse probability of censoring, which removes the dependence on the censoring distribution. Python implementations are provided by lifelines [32] and scikit-survival [33].

2.5 Rule Learning for Survival Data

LR-Rules [34] is a sequential-covering algorithm for survival data. It grows each rule by adding the condition that maximises the log-rank separation between covered and uncovered patients. Each rule's conclusion is the Kaplan–Meier curve of the patients it covers. RuleKit [10] packages LR-Rules, together with classification and regression rule learners, as a Java library with a Python interface. Survival rules are readable statements such as "IF nodal stage is N3 AND aneuploidy is below 26.5 THEN the progression-free survival curve is …", which makes them a natural symbolic component.

2.6 Statistical Relational Learning: Markov Logic and Soft Logic

Markov logic networks (MLNs) [38] attach a real weight w_k to each first-order formula F_k and define a log-linear distribution over possible worlds, $P(x) \propto \exp\big(\sum_k w_k\, n_k(x)\big)$, where $n_k(x)$ is the number of true groundings of F_k. A formula with a large weight is a soft constraint: worlds that violate it are less probable but not impossible. MLN learning has two parts: structure learning finds the formulas [39], and weight learning fits w, usually by maximising a pseudo-likelihood.

Probabilistic soft logic (PSL) [40] relaxes truth values to $[0, 1]$ with the Łukasiewicz t-norm. A weighted rule $w : B \Rightarrow H$ becomes the hinge potential $w \max(0, B - H)^q$, its distance to satisfaction. The resulting hinge-loss Markov

random field has convex MAP inference, so PSL scales to collective reasoning over many interrelated entities, for example propagating a label over a similarity graph.

2.7 Reinforcement Learning for Rules and Paths

Policy-gradient reinforcement learning [41] optimises a stochastic policy from scalar rewards without a differentiable objective. In knowledge-graph reasoning, DeepPath [42] and MINERVA [43] train agents to walk relation paths, and RNNLogic [44] trains a recurrent rule generator jointly with a reasoning predictor that scores its rules. We follow the generator idea with a gated recurrent unit [45] that writes conjunctive survival rules, rewarded by log-rank separation, and we use its output as the structure learner of a survival MLN.

2.8 Neuro-Symbolic AI

Garcez and Lamb [7] describe neuro-symbolic AI as the integration of neural learning with symbolic knowledge representation and reasoning, so that systems can both learn from data and explain themselves with explicit knowledge. In biomedicine this typically means grounding neural predictions in curated knowledge graphs and rules. Our design follows that pattern. The neural component scores knowledge-graph paths, and the symbolic component, an RL-structured survival Markov logic network, states cohort-level regularities as weighted first-order formulas over the same knowledge graph. Each path is scored by both components.

2.9 Gene-Expression Prognosis Cohorts

GSE2034 [35] (286 lymph-node-negative patients, Affymetrix arrays) and GSE2990 [36] (189 patients with relapse-free survival) are classic external breast cancer prognosis cohorts. We use them as an external benchmark for the graph models.

Chapter 3

Proposed Work

Figure 1 gives an overview of the system. Patient graphs are built from TCGA and fused with a filtered PrimeKG neighbourhood. A survival GNN and four survival-aware logic reasoners run on the same cohort and the same folds, and the best reasoner, an RL-structured survival Markov logic network, becomes the logic engine. The GNN and logic risks are fused into one Cox model. Every patient receives a report that combines the prediction with knowledge-graph paths, each scored by GNN importance and by the weighted logic formulas that hold for the patient.

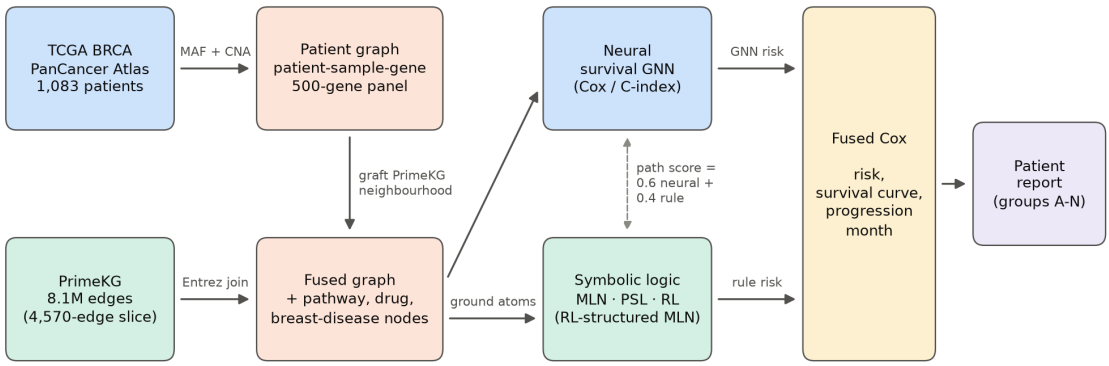


Figure 1: Overview of the neuro-symbolic progression system.

3.1 Data

The study is the cBioPortal `brca_tcga_pan_can_atlas_2018` bundle. We verified every data file against the official SHA-256 checksums. Four tables are used:

- `data_clinical_patient.txt`: 1,084 patients, 38 columns, including the PFS status and months;
- `data_clinical_sample.txt`: 1,084 primary tumour samples, 19 columns;
- `data_mutations.txt`: 126,252 MAF mutation records with 111 annotation columns;
- `data_cna.txt`: discrete GISTIC calls for 25,128 genes across 1,070 samples.

A streamed audit of all 23 tables in the bundle records their schemas and missingness. The expression, methylation, protein, and structural-variant modalities were inspected and deliberately excluded to keep the graph interpretable.

Leakage policy. Outcome columns (OS, DSS, DFS, and PFS status and months, the new-tumour-event flag, and the cancer status at last contact) are never features. A guard function rejects any feature whose name matches an outcome column, and tests assert it.

3.2 Patient Graph Construction

Each patient is a `networkx` `MultiDiGraph` with the schema Patient $\rightarrow$ Sample $\rightarrow$ Gene (Figure 2):

- **Patient** nodes carry age, sex, PAM50 subtype, AJCC stage, and pathological M stage.
- **Sample** nodes carry the tumour type, OncoTree code, aneuploidy score, and a raw MAF record count.
- **Gene** nodes are keyed by HGNC symbol and Entrez identifier. The identifier pair avoids merging the duplicated symbol rows of the CNA matrix.

- `has_mutation` edges preserve each MAF record (variant class, protein change, coordinates, alleles, caller filters). `has_CNA` edges carry the GISTIC value.

Exact duplicate CNA calls are deduplicated. Contradictory calls for the same gene and sample are withheld and logged rather than resolved arbitrarily. Every graph is validated before export.

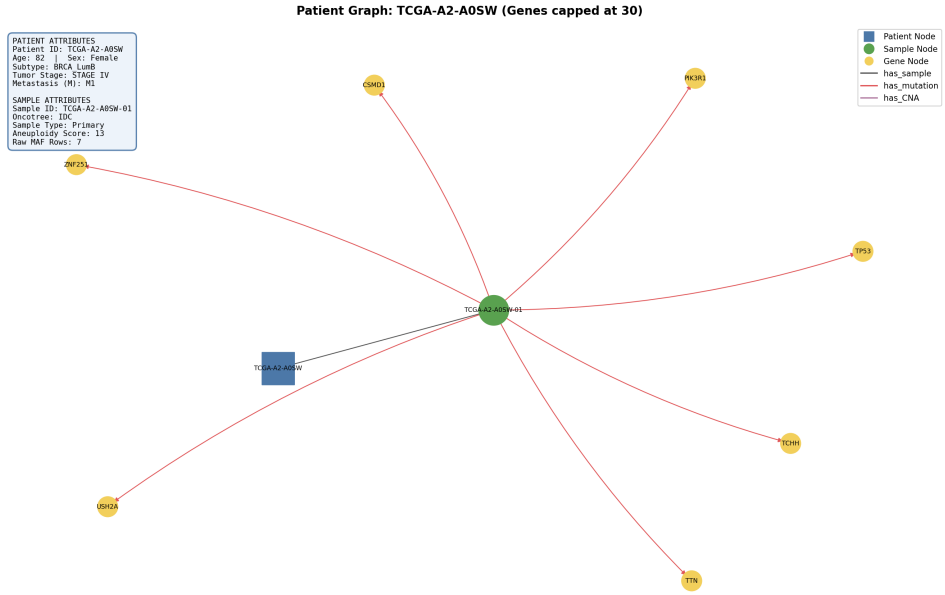


Figure 2: A Milestone 1 patient graph (TCGA-A2-A0SW): one patient, one tumour sample, and its mutated genes.

3.3 Gene Panel

To bound graph size, genes are restricted to a 500-gene panel ranked by an integrated alteration score: the number of samples with a mutation plus the number with an extreme CNA (-2 or $+2$). Twenty-seven literature breast cancer drivers are always included: TP53, PIK3CA, GATA3, CDH1, ERBB2, PTEN, MAP3K1, BRCA1, BRCA2, RB1, CCND1, MYC, ATM, ESR1, AKT1, MAP2K4, KMT2C, ARID1A, NF1, EGFR, FGFR1, FOXA1, CBFB, RUNX1, PIK3R1, SF3B1, and

BAP1. For modelling, CNA evidence is limited to homozygous deletion and high-level amplification.

3.4 Fusion with PrimeKG

PrimeKG's 8,100,498 edges are streamed once and filtered to a 4,570-edge panel slice that keeps four relations:

- `protein_protein` edges between two panel genes (772);
- `pathway_protein` edges from panel genes to Reactome pathways (2,410);
- `drug_protein` edges from panel genes to DrugBank drugs (858);
- `disease_protein` edges to diseases whose name contains "breast" or "mammary" (530).

Patient genes join PrimeKG proteins on Entrez identifier. The fused graph (Figure 3) adds, for each altered gene, its panel protein partners, its pathways, drugs, and breast-disease links, and the unaltered panel genes that share a pathway with it (two hops). Every relation is also stored in reverse so that three message-passing layers can carry pathway context back to the patient node. Over the cohort the median fused graph has 632 nodes and 2,526 directed edges.

Node features have one width: a node-type one-hot, a 19-field clinical block on the patient node, a five-field alteration block on gene nodes (mutated, amplified, deleted, log(1 + mutation count), known driver), and a trainable embedding for every gene, pathway, drug, and disease.

3.5 Progression Labels

The target is the progression-free interval (`PFS_STATUS`, `PFS_MONTHS`). 1,083 patients have a usable status, and 145 of them have a recorded progression. A censored patient was not seen to progress during follow-up. Treating censoring as "never progresses" biases a binary model, so two endpoints are used:

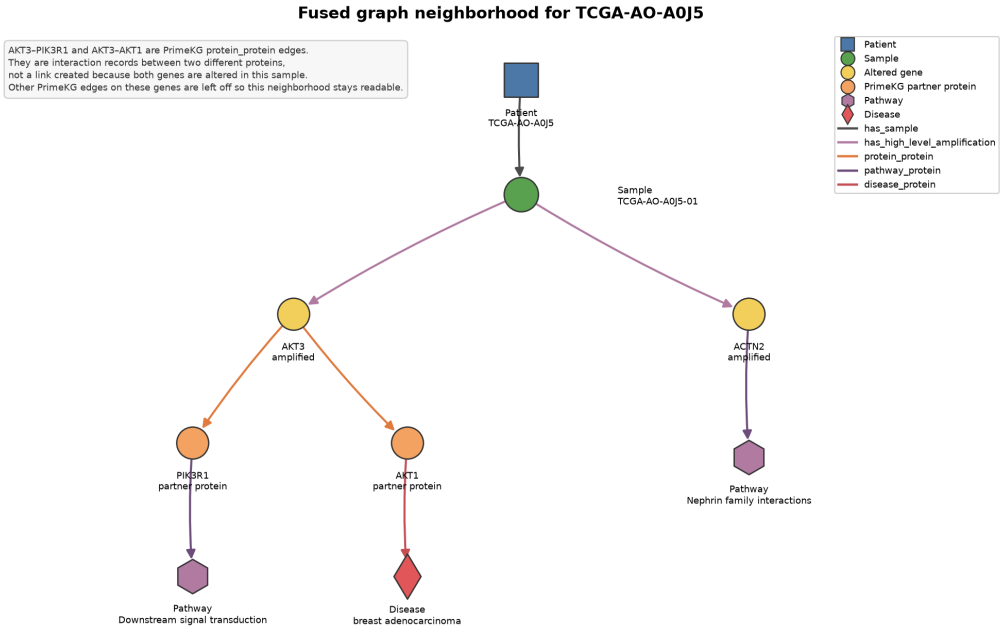


Figure 3: Part of a fused graph: altered genes of one patient connected to PrimeKG partner proteins, pathways, and a breast-disease node.

- **36-month landmark** (`pfs_36m`): label 1 for progression at or before 36 months and 0 for follow-up beyond 36 months. Patients censored before 36 months are dropped, which leaves 498 patients and 89 events.
- **Full follow-up** (`pfs`): the time and event pair for all 1,083 patients, used by the survival objectives and by the neuro-symbolic model.

The PFS months define the target and the comparable pairs. They are never model inputs.

3.6 Graph Neural Network Models

Seven architectures share one design: three message-passing layers of width 64, dropout 0.3, and a readout that concatenates the patient node with mean- and max-pooling over the patient and altered genes. A linear clinical head is added to the graph score through a gate that starts near zero, so training begins from the

clinical score. The architectures are GCN, GraphSAGE, GATv2 (four heads), R-GCN and RGAT over the eight relation types, GIN (a control that ignores relations), and a graph transformer that reads a learned relation embedding as an edge feature. A multilayer perceptron over the clinical block and a 500-gene alteration multi-hot, and a clinical logistic regression, are the non-graph baselines.

3.6.1 Training Objectives

- **Weighted binary cross-entropy** on the landmark label.
- **Differentiable C-index**: for comparable pairs (i, j) with $t_i < t_j$ and $\delta_i = 1$, minimise $1 - \frac{1}{|P|} \sum_{(i,j)\in P} \sigma\big(\frac{r_i - r_j}{\tau}\big)$ over each mini-batch.
- **Cox partial likelihood** (Breslow ties):

$$\mathcal{L} = -\frac{1}{N_e} \sum_{i:\,\delta_i=1} \Big(r_i - \log \sum_{j:\,t_j \geq t_i} e^{r_j} \Big), \tag{1}$$

  where r_i is the model's log-risk for patient i.

3.7 Patient-Conditioned Graph Structure Learning

A fixed two-hop neighbourhood may include PrimeKG edges that are irrelevant to a given patient. The graph-structure-learning (GSL) variant retrieves up to three hops and learns a weight for every PrimeKG edge from the edge's endpoints, its relation, and the patient's clinical context. In the hop-prior version, edges within the fixed two hops start at weight 1, so training starts from the fixed graph. The third hop sits behind a patient-specific gate that starts closed (about 0.12) and must cross 0.5 to contribute. The learned weight multiplies messages in GCN, GraphSAGE, and GATv2. The relational convolutions in PyTorch Geometric accept no edge weight, so GSL with them is refused rather than silently ignored.

3.8 Neuro-Symbolic Progression Model

The final system has four stages, all run over one cohort (1,083 patients) and one set of stratified five-fold splits. Every prediction, rule, and path score for a patient therefore comes from models trained on the four folds that exclude that patient.

3.8.1 Neural Component: Survival GNN

GCN, GraphSAGE, and R-GCN are trained on the fused graphs with the Cox objective, and GraphSAGE also with the C-index objective, on the full follow-up. Mini-batches of 128 graphs are used for the Cox loss so that each batch holds enough events for its risk sets. Early stopping uses the validation C-index. The best model by fold-mean C-index supplies the neural risk. For every patient, GNNExplainer [6], run on the fold model that held that patient out, supplies a soft importance for every edge of the patient's graph.

3.8.2 Symbolic Baseline: Predicates and RuleKit Rules

Each patient is described by 188 readable predicates:

- **Clinical (10)**: age, T stage, N stage, M stage, AJCC stage, PAM50 subtype, histology (ductal, lobular, other), radiation therapy, aneuploidy score, and tumour mutational burden.
- **Genomic (58)**: mutations of the 27 drivers, driver copy-number changes, amplicons, and the number of altered panel genes. Co-amplified genes whose patient sets overlap with Jaccard index ≥ 0.8 are merged into one predicate named after a driver, for example "MYC amplified (+27 co-altered genes)". Mutations of non-driver genes, mostly long passenger genes such as TTN or RYR2, are excluded.
- **Knowledge-graph (120)**: "an altered gene of this patient is annotated to Reactome pathway P" (116) or "…is linked to breast disease D" (4). Predicates are kept when 5–60% of patients carry them, and near-duplicates are merged.

Predicate selection uses prevalence only, never the outcome. Binary predicates enter RuleKit as "yes" or missing, so a rule can say "TP53 mutated" but cannot be built from the absence of a rare mutation.

As a baseline, RuleKit's LR-Rules [34, 10] is fitted on each training fold (minimum coverage 5% of new patients, at most three conditions per rule). A new patient's prediction is the average Kaplan–Meier curve of the rules that cover them. The rule risk is the negative restricted mean survival time up to 120 months:

$$r_i^{\text{rule}} = -\int_0^{120} \hat{S}_i^{\text{rule}}(t)\, dt. \tag{2}$$

3.8.3 Knowledge-Graph Fact Evidence

For every PrimeKG pathway, drug, and disease node v, the fact f_v is "the patient has an altered gene directly annotated to v". On each training fold a vectorised two-group log-rank test compares carriers with non-carriers, giving an observed/expected hazard-ratio estimate and a z statistic z_v. Facts with fewer than 15 carriers are skipped. In total 1,053 facts receive evidence.

3.8.4 Logic Reasoners: Beyond IF–THEN Rules

RuleKit's crisp rules cover few patients, carry no weights, and rarely use the knowledge graph. We therefore build four survival-aware reasoners over a shared set of first-order *ground atoms* and compare them on the same folds.

Ground atoms. Every atom is a closed formula about a patient p, and its value $n(p)$ is its number of true groundings. There are 26 clinical atoms such as $\mathit{NStage}(p, \text{N3})$ or $\mathit{Age}(p) \geq 65$, 57 genomic atoms such as $\mathit{Mutated}(p, \text{TP53})$, and 339 relational PrimeKG atoms

$$n_w(p) = \big|\{\, g : \mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, w) \,\}\big|, \tag{3}$$

and likewise with $\mathit{TargetedBy}(g, d)$ for drugs and $\mathit{LinkedTo}(g, d)$ for breast diseases. Atoms are kept at 3–90% prevalence, which never uses the outcome. Atoms with identical carrier sets are merged, and the merged atom keeps every PrimeKG node it stands for.

Survival Markov logic network (Cox-MLN). Every formula concludes progression, $w_k : \mathit{Body}_k(p) \Rightarrow \mathit{Progress}(p)$. The bodies are every atom plus up to 150 clinical/genomic $\times$ PrimeKG conjunctions mined on the training fold. With fully observed evidence and one query atom per patient, MLN inference factorises patient by patient into the score $\eta(p) = \sum_k w_k \log\big(1 + n_k(p)\big)$, where $\log(1+n)$ dampens patients with many co-amplified genes in one pathway. The usual conditional likelihood would treat censored patients as non-progressors. We instead learn the weights by maximising the elastic-net-penalised Cox partial likelihood [46, 47]

$$\max_{w} \sum_{i:\,\delta_i=1} \Big(\eta_i - \log \sum_{j:\,t_j \geq t_i} e^{\eta_j}\Big) \;-\; \lambda\Big(\alpha\|w\|_1 + \tfrac{1-\alpha}{2}\|w\|_2^2\Big), \tag{4}$$

with $\alpha = 0.5$ and λ chosen by inner three-fold C-index. The ℓ_1 term performs structure learning: formulas whose weight reaches zero are pruned.

Survival probabilistic soft logic (S-PSL). The query atom $\mathit{HighRisk}(p) = h_p \in [0,1]$ is inferred from the program

$$\begin{aligned}
&w_k^{+} : \mathit{Fact}_k(p) \Rightarrow \mathit{HighRisk}(p),\\
&w_k^{-} : \mathit{Fact}_k(p) \Rightarrow \neg\mathit{HighRisk}(p),\\
&s^{+} : \mathit{Similar}(p,q) \wedge \mathit{HighRisk}(q) \Rightarrow \mathit{HighRisk}(p),\\
&s^{-} : \mathit{Similar}(p,q) \wedge \neg\mathit{HighRisk}(q) \Rightarrow \neg\mathit{HighRisk}(p),
\end{aligned} \tag{5}$$

plus a prior $\neg\mathit{HighRisk}(p)$. Facts are the 40 strongest atoms on the training fold, with PrimeKG counts softened to $\min(1, n/2)$. $\mathit{Similar}(p,q)$ links each patient to their ten most similar training patients by cosine similarity of atoms. For those neighbours h_q is observed: it is a censoring-aware soft target, the Kaplan–Meier probability of progression by 36 months given the patient's follow-up. MAP inference minimises the squared-hinge energy by 40 projected-gradient steps. Inference is unrolled, so the rule weights are learned end to end by soft-label cross-entropy plus a Cox term.

Reinforcement-learned rules (RL-Rules). A GRU policy [45] writes a rule one atom at a time (up to three atoms, or STOP), masking atoms of an already chosen clinical variable. It is trained with REINFORCE [41], a moving-average baseline,

and an entropy bonus over 160 batches of 256 episodes per fold. The training fold is split into a 70% search part, halved into A and B, and a 30% validation part. The reward is a *stability reward*:

$$R(\text{rule}) = \mathbb{1}\big[\operatorname{sign} z_A = \operatorname{sign} z_B\big]\,\frac{\min\big(|z_A|, |z_B|, 5\big)}{5} \;-\; 0.03\,(\text{length} - 1), \qquad (6)$$

where z_A and z_B are the log-rank statistics of the covered patients against the rest on the two halves. Rules covering fewer than 4% or more than 90% of the fold earn nothing. Because tens of thousands of distinct rules are sampled, the best training rewards are a multiple-testing artefact. Only rules whose separation replicates on the untouched 30% (same sign, $|z| \geq 1$) are kept, and a ridge Cox model on their activations gives the risk.

RL-structured survival MLN (engine). The RL search proposes up to 150 replicated multi-atom formulas. They are added to the Cox-MLN's candidates, and the Cox-MLN learns every weight. Reinforcement learning does the MLN's structure learning and the survival likelihood does its weight learning, combining two of the three families.

Choosing the engine. The reasoners are compared by fold-mean C-index and the one-standard-error rule [48]. Among reasoners within one standard error of the best, the earliest in the preference order RL-structured MLN > MLN > PSL > RL rules > RuleKit is the engine, which puts the richest reasoning first.

3.8.5 Fusion and Model Selection

The fused model is a ridge-penalised Cox model on the 19-field clinical block, the out-of-fold GNN risk, and the out-of-fold risk of the logic engine. This is cross-validated stacking [37]. It is compared with clinical Cox models (19 fields and an extended 27-column design), the neural and symbolic risks alone, and a random survival forest [28]. The report model is chosen by the one-standard-error rule [48]. Let $\bar{C}^*$ be the best fold-mean C-index and s^* its standard deviation over the $k = 5$ folds. Every model with $\bar{C} \geq \bar{C}^* - s^*/\sqrt{k}$ is statistically tied with the best, and among those the most explainable model, the neuro-symbolic one, is chosen.

3.8.6 Estimating the Month of Progression

Most patients never progress, so the plain median of a Cox survival curve is usually undefined. The quantity reported is the time to progression *given* that progression occurs. Three families of estimators are compared on held-out progressors:

- the conditional median of the fused survival curve, the month at which $F(t)/F(\tau) = 1/2$ with $F = 1 - S$, for $\tau \in \{60, 120, 180\}$ months;
- log-normal and log-logistic AFT models [27] fitted on the training-fold progressors only, using the extended clinical design, the GNN risk, and the logic-engine risk;
- a constant baseline, the median progression month of the training-fold progressors.

The estimator with the lowest mean absolute error is used. Its interquartile range is reported with it, and the estimate is mapped to one of four windows.

3.8.7 Neuro-Symbolic Path Scoring and Top-K Selection

A candidate explanation path is a walk over stored edges of the patient's fused graph: patient → sample → altered gene → (protein partner) → pathway, drug, or breast disease. Up to 60 candidates are enumerated per patient. Let $\bar{m}(p)$ be the mean GNNExplainer importance of path p's edges and $\mathrm{pct}(\cdot)$ its percentile among all candidate paths in the cohort. Let $\mathcal{F}(p)$ be the logic-engine formulas, from the fold that held the patient out, that the path *grounds*: formulas that fire for the patient and have an atom that mentions the path's end node or altered gene, or a relational PrimeKG atom $\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, w)$ whose gene g lies on the path. With S the 95th percentile of $|\sum_f w_f|$ over all rule-matched paths,

$$\mathrm{neural}(p) = \mathrm{pct}\left(\bar{m}(p)\right), \tag{7}$$

$$\mathrm{rule}(p) = \min\Big(\big|\textstyle\sum_{f \in \mathcal{F}(p)} w_f\big| \,/\, S,\ 1\Big), \tag{8}$$

$$\mathrm{NS}(p) = 0.6\,\mathrm{neural}(p) + 0.4\,\mathrm{rule}(p). \tag{9}$$

The sign of $\sum_f w_f$ gives the direction (faster or slower progression). The log-rank evidence of the end fact (Section 3.8.3) is printed beside the path as cohort context, but no longer drives the score. A path is *prominent* when its NS score is in the top 20% of all candidates in the cohort. A report shows $K = \min(5, \text{number of prominent paths})$. At most two paths may pass through the same gene, and each gene route and each end node appears once. When fewer than three paths are prominent, the best remaining paths are added and marked as non-prominent.

3.9 Report Groups

Reports are grouped by the *recorded* progression month, so a group never mixes true progressors with patients who were not seen to progress (Table 1). Only 12 of the 145 progressions happen within six months, so the first group is "within 12 months".

Table 1: Report groups by recorded progression-free survival.

Group	Recorded progression	Patients
A	≤ 12 months	32
B	12–24 months	30
C	24–36 months	27
D	> 36 months	56
N	no progression recorded (censored)	938

3.10 Patient Report Design

Each report is an A4 PDF of two to three pages, produced with ReportLab and Matplotlib. It contains:

1. **Patient and tumour record**: demographics, PAM50 subtype, AJCC stage and TNM, histology, therapy history, aneuploidy, TMB, and MSI score.

2. **Progression assessment**: the risk percentile and tier with a gauge, the probability of progression by 12, 24, 36, and 60 months, the estimated progression

month and window, the recorded outcome with a tick or cross against the estimate, and the patient's survival curve against the cohort Kaplan–Meier curve.

3. **Neuro-symbolic reasoning**: the top-K paths drawn as chains with their neural and symbolic score split, a table of the rule evidence behind each path, and the weighted logic formulas that fire for the patient, with their weight, direction, source (RL-proposed, mined, or single atom), support, and the Kaplan–Meier curve of the training patients for whom the strongest formula holds.

4. **Molecular profile and knowledge-graph context**: altered genes with protein change, PrimeKG links, and GNN importance; PrimeKG drug links (research context only); and the five most similar patients by cosine similarity of their predicates, with their recorded outcomes.

Each progression group also gets a summary page (group statistics, frequent path end-points, frequent rules, common driver alterations, and a patient index) and a combined PDF. A cohort overview compares all models.

Chapter 4

Experimental Setup and Results Analysis

4.1 Experimental Setup

Hardware and software. All experiments ran on an Apple Silicon laptop (8 CPU cores, 8 GB RAM) without a GPU. Software: Python 3.13, PyTorch 2.14, PyTorch Geometric 2.8, scikit-learn 1.9, lifelines 0.30, scikit-survival 0.28, RuleKit 2.1.24 on Java 22, and ReportLab 5.0. The full neuro-symbolic pipeline runs in about 50 minutes, and the 1,083 reports build in about 5 minutes on six processes.

Protocol. All evaluation is stratified five-fold cross-validation with seed 42. Within each training fold, 15% is held out for early stopping. Clinical standardisation is fitted on training folds only. Graph models train for up to 30–40 epochs with patience 6–8, using Adam (learning rate 10^{-3}, weight decay 10^{-4}).

Metrics. AUROC and AUPRC for the landmark label; Harrell's C (pooled, and the mean $\pm$ standard deviation over folds); Uno's C truncated at 120 months; time-dependent AUC at 12, 36, and 60 months; and the integrated Brier score up to 120 months. For the month of progression: mean and median absolute error, the share of estimates within 12 months, window accuracy, and interquartile-range coverage.

4.2 Milestone 1: Patient Graphs

Graphs were built and validated for every patient. A raw-PFS binary GraphSAGE model on all 1,083 patients reached only AUROC 0.542 and AUPRC 0.216, close to chance. Censoring makes "no recorded progression" an unreliable negative label, which motivated the landmark endpoint.

4.3 Milestone 2: PrimeKG Fusion and the 36-Month Benchmark

Table 2 shows the first fused-graph benchmark on the 36-month landmark (498 patients, 89 events; the prevalence, which is the AUPRC of a random ranking, is 0.179). The clinical logistic regression is the best model. GCN is the best graph model and comes close without beating it. The gene multi-hot MLP is the weakest, because the wide alteration vector adds noise. Figure 4 shows the out-of-fold ROC curves of the graph models.

Table 2: 36-month progression, five-fold mean ± standard deviation.

Model	AUROC	AUPRC
Clinical logistic regression	0.703 ± 0.081	0.463 ± 0.133
GCN	0.680 ± 0.076	0.396 ± 0.091
GraphSAGE	0.650 ± 0.045	0.379 ± 0.091
GATv2	0.631 ± 0.046	0.355 ± 0.060
MLP (clinical + gene multi-hot)	0.562 ± 0.098	0.262 ± 0.064

Choice of horizon. Figure 5 sweeps the landmark from 12 to 60 months with the clinical model. Short horizons have too few events (32 at 12 months). Long horizons drop too many censored patients (only 344 remain at 60 months). The 36-month horizon balances the two.

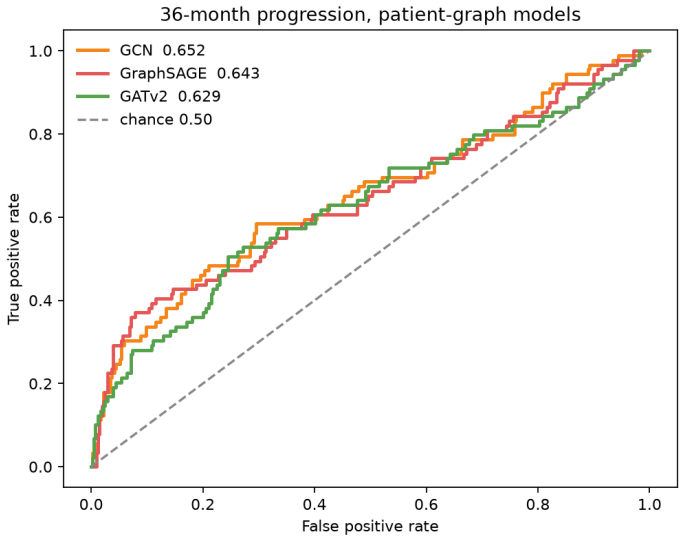


Figure 4: Out-of-fold ROC curves of the graph models on the 36-month endpoint.

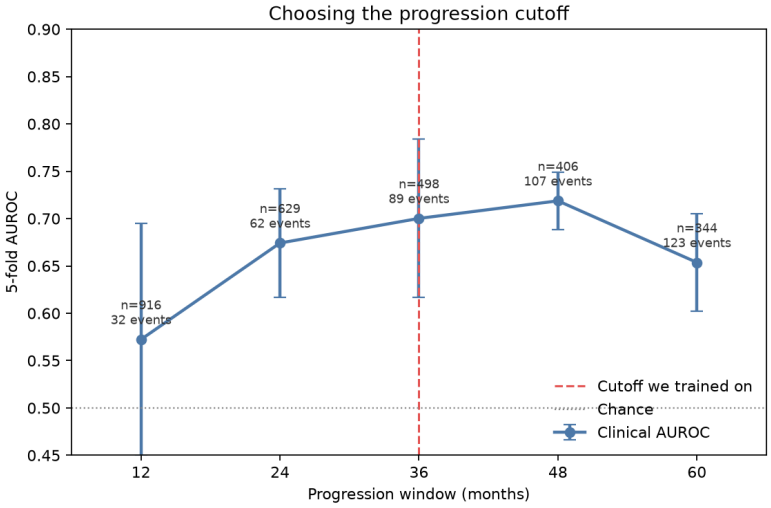


Figure 5: Clinical AUROC as a function of the progression window.

4.4 Concordance Training and the Architecture Zoo

Table 3 compares the seven architectures under binary cross-entropy and under the C-index objective on the same folds. RGAT and GCN are the strongest graph models. GIN, which ignores relation types, is the weakest, so typing the edges matters. The C-index objective does not improve discrimination over cross-entropy. No graph model beats the clinical baseline (AUROC 0.703, C-index 0.682), and fold standard deviations of 0.05–0.15 exceed most of the differences.

Table 3: 36-month endpoint: AUROC, AUPRC, and Harrell's C (five-fold means) under two objectives.

	Cross-entropy			C-index objective		
Model	AUROC	AUPRC	C	AUROC	AUPRC	C
RGAT	0.694	0.412	0.672	0.678	0.389	0.658
GCN	0.688	0.415	0.665	0.658	0.379	0.639
GraphSAGE	0.679	0.406	0.658	0.660	0.393	0.643
GATv2	0.638	0.366	0.626	0.640	0.366	0.628
R-GCN	0.627	0.352	0.616	0.602	0.343	0.597
Transformer	0.613	0.354	0.607	0.627	0.376	0.617
GIN	0.590	0.320	0.581	0.558	0.263	0.555
MLP	0.570	0.274	0.562	0.566	0.263	0.560

4.5 Graph Structure Learning

Figure 6 compares the fixed two-hop graph with two GSL variants. The first variant used free edge weights and let the scorer read the hop index. It closed the entire second hop (mean weight 0.24, and none of 5,745 hop-2 edges above 0.5), and GCN's AUPRC fell from 0.396 to 0.316. The hop-prior variant starts from the fixed graph and gates only the third hop. Its run reported higher means (AUROC about 0.70 and AUPRC about 0.46 for all three models), but within the fold standard deviations.

That run's directory was not kept, so those bars are transcribed from its report and cannot be recomputed.

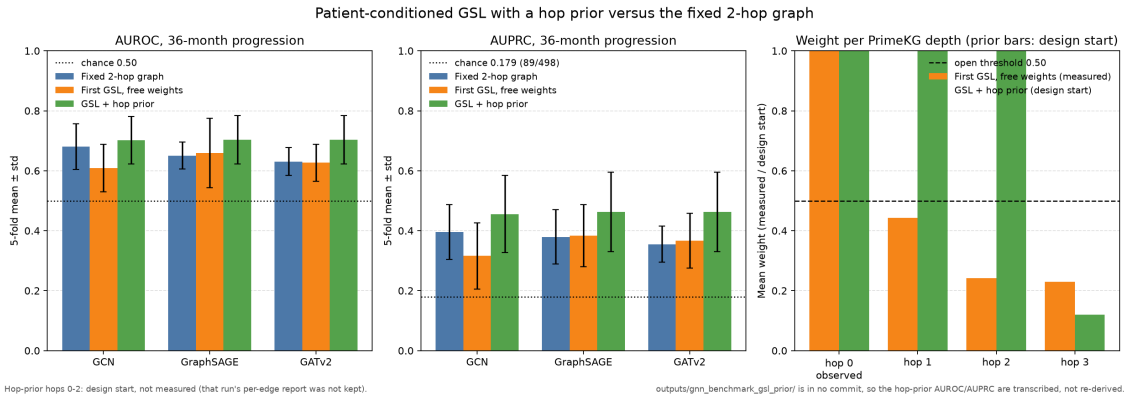


Figure 6: Graph structure learning against the fixed two-hop graph on the 36-month endpoint.

4.6 External Benchmark on GSE2034

On the GSE2034 microarray cohort (286 patients, bone-relapse label, a 4,712-gene PrimeKG-aligned expression graph), logistic regression reached AUROC 0.722 ± 0.078 and AUPRC 0.593. The random forest reached 0.682, GraphSAGE 0.582, GCN 0.568, and a majority baseline 0.500. As on TCGA, the expression graph did not lift the GNNs above a linear baseline. A Cox model on principal components of the TCGA clinical block reached a test C-index of 0.659 ± 0.019.

4.7 Neuro-Symbolic Model

4.7.1 Cohort

Figure 7 shows when the 145 recorded progressions occur. The median is 26 months, and progressions continue past 10 years.

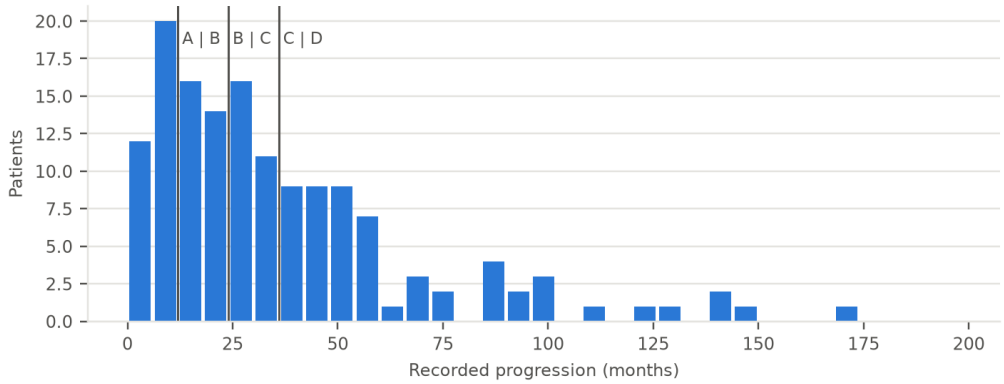


Figure 7: Recorded progression months, with the report group boundaries at 12, 24, and 36 months.

4.7.2 Discrimination and Model Selection

Table 4 and Figure 8 compare all models on the full follow-up, out-of-fold, on the same five folds.

The best fold-mean C-index is 0.675 (clinical Cox), with one standard error of 0.016. Four models clear the bar of 0.659 and are statistically tied with it. Among them the neuro-symbolic Cox model is chosen because it is the one whose prediction can be explained through paths and rules.

Two findings matter for interpretation:

1. **Saturation.** GNNs (0.59–0.62), logic reasoners (0.56–0.63), forests (0.55), and Cox models (0.662–0.675) all stop near the same ceiling. Without the clinical block, the neural and logic risks reach 0.642 together, better than either alone (0.622 and 0.628). Every Cox model that includes the clinical block reaches 0.662–0.675.

2. **Explanation, not accuracy.** In every fold the fused model gives the GNN and logic terms hazard ratios of 0.94–1.09 per standard deviation with $p > 0.28$. Once the clinical block is present they add no independent prognostic information. The reports state this explicitly.

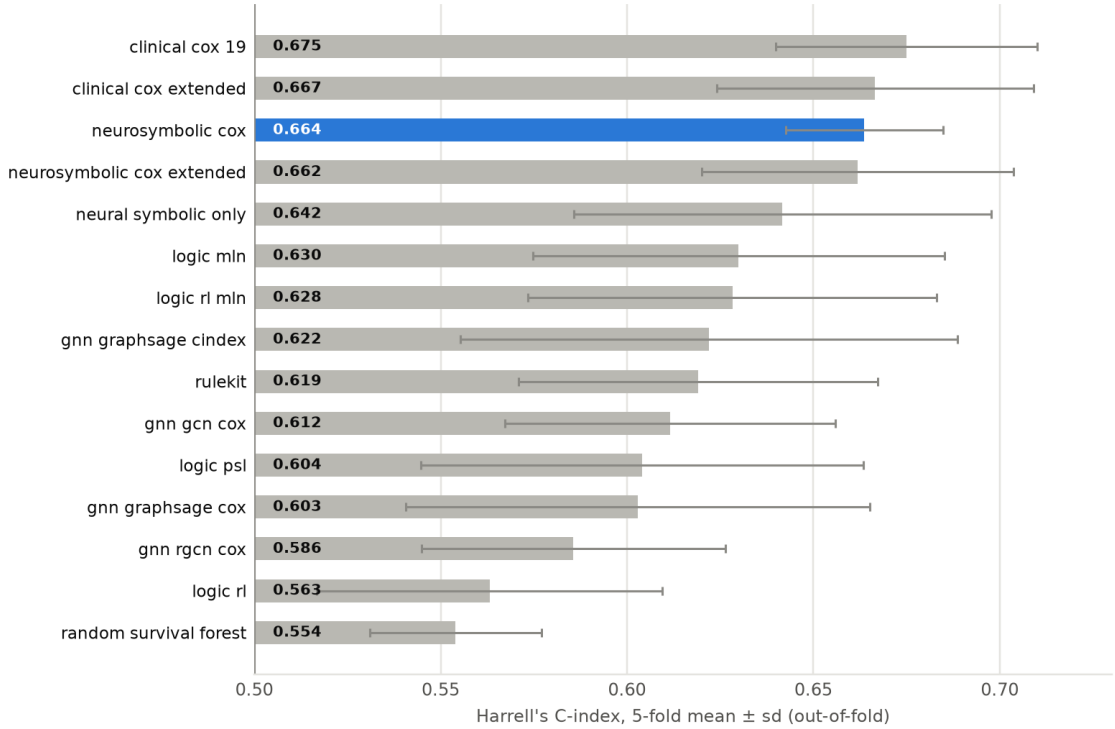


Figure 8: Fold-mean Harrell's C-index of every model. The report model is highlighted.

Table 4: Full PFS follow-up (1,083 patients, 145 events). Harrell's C is the five-fold mean ± standard deviation. Uno's C is truncated at 120 months, and IBS is the integrated Brier score to 120 months (lower is better).

Model	Harrell's C	Uno's C	IBS
Clinical Cox (19-field block)	0.675 ± 0.035	0.616	0.134
Clinical Cox (extended)	0.667 ± 0.042	0.657	0.135
Neuro-symbolic Cox (clin. + GNN + logic)	$\mathbf{0.664 \pm 0.021}$	0.616	0.134
Neuro-symbolic Cox (ext. clinical)	0.662 ± 0.042	0.655	0.135
GNN + logic, no clinical	0.642 ± 0.056	0.606	0.139
Survival MLN (logic alone)	0.630 ± 0.055	0.617	–
RL-structured MLN (logic engine alone)	0.628 ± 0.055	0.615	–
GraphSAGE (C-index objective)	0.622 ± 0.067	0.573	–
RuleKit survival rules	0.619 ± 0.048	0.623	–
GCN (Cox objective)	0.612 ± 0.044	0.549	–
GraphSAGE (Cox objective)	0.603 ± 0.062	0.556	–
R-GCN (Cox objective)	0.586 ± 0.041	0.590	–
Survival PSL	0.604 ± 0.059	0.571	–
RL rules	0.563 ± 0.046	0.570	–
Random survival forest	0.554 ± 0.023	0.559	0.145

4.7.3 Symbolic Reasoners Compared

Table 5 and Figure 9 compare the four logic reasoners with the RuleKit baseline on the same folds.

Table 5: Symbolic reasoners on the full follow-up: five-fold Harrell's C (mean ± sd), Uno's C, learned formulas per fold, and the share of formulas that reason over a PrimeKG relation.

Reasoner	Harrell's C	Uno's C	Formulas/fold	PrimeKG
Survival MLN (Cox-MLN)	0.630 ± 0.055	0.617	32.8	63%
RL-structured survival MLN (engine)	$\mathbf{0.628 \pm 0.055}$	0.615	35.2	**69%**
RuleKit LR-Rules (baseline)	0.619 ± 0.048	0.623	7.8	10%
Survival PSL (collective)	0.604 ± 0.059	0.571	40.0	63%
RL rules (policy gradient)	0.563 ± 0.046	0.570	25.6	91%

What the comparison shows.

1. **Markov logic leads, but within noise.** Both MLN variants (0.630 and 0.628) are ahead of RuleKit (0.619), PSL (0.604), and RL rules alone (0.563).

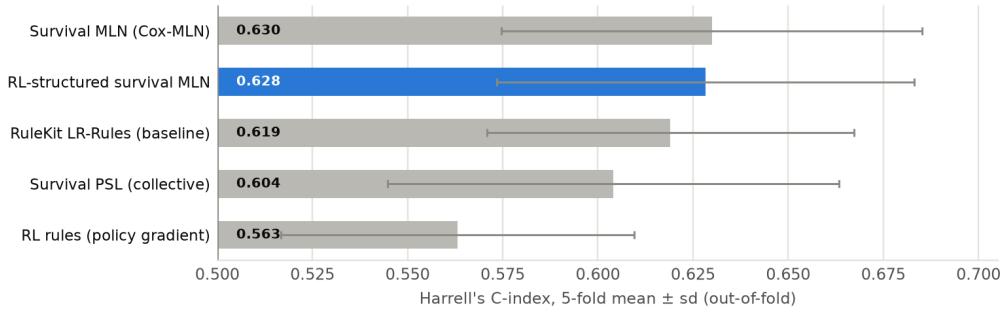


Figure 9: Fold-mean Harrell's C of the symbolic reasoners. The engine is highlighted.

Their margin over RuleKit is smaller than one fold standard deviation, so the gain is not in discrimination. Under the one-standard-error rule (bar 0.605) MLN, RL-structured MLN, and RuleKit are tied, and the preference order selects the RL-structured MLN.

2. **The gain is in the reasoning.** 69% of the engine's formulas reason over PrimeKG relations, against 10% of RuleKit's rules. The formulas are weighted, combinable, and multi-hop. Across the folds the engine kept 176 formulas: 51 single atoms, 81 mined clinical × PrimeKG pairs, and 44 multi-atom formulas proposed by reinforcement learning.

3. **Collective inference did not help.** PSL learned a collective weight of about 0.01: in this cohort, similar patients carry no outcome signal beyond their own facts.

4. **RL search overfits on its own.** The policy sampled 20,000–27,000 distinct rules per fold. Even with the replication filter, the rules it ranks highest do not transfer well (0.563). RL is useful as the structure learner of the MLN, where the Cox likelihood and its ℓ_1 penalty decide which proposals carry weight.

Learned formulas. Clinical conditions still carry the largest single weights, for example $\mathit{NStage}(p, \mathrm{N0}) \Rightarrow \mathit{Progress}(p)$ (weight -0.32 per SD) and $\mathit{TStage}(p, \mathrm{T4})$

(+0.13). Typical PrimeKG formulas are a mined pair (+0.15, 43 patients),

$$\mathit{NStage}(p, \mathrm{N3}) \wedge \mathit{Altered}(p, g) \\ \wedge \mathit{LinkedTo}(g, \text{breast ductal adenocarcinoma}) \Rightarrow \mathit{Progress}(p),$$

and an RL-proposed multi-hop formula (-0.14, 93 patients, a protective formula; Figure 10),

$$\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{VEGFA-VEGFR2}) \\ \wedge \mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{Ion homeostasis}) \Rightarrow \mathit{Progress}(p).$$

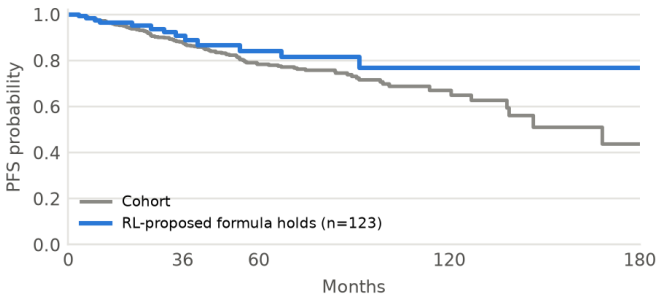


Figure 10: Kaplan–Meier curve of the patients for whom the RL-proposed formula $\mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{VEGFA-VEGFR2}) \wedge \mathit{Altered}(p, g) \wedge \mathit{InPathway}(g, \text{Ion homeostasis})$ holds, against the cohort.

4.7.4 Estimating the Month of Progression

Table 6 scores the estimators on the 145 held-out progressors. The conditional medians of the Cox curve are poor, because proportional hazards fix the shape of the curve. The AFT models fitted on progressors are best. The chosen log-logistic AFT model is within 12 months of the recorded month for 48% of progressors (37% for the constant baseline) and in the recorded window for 33% (19% for the baseline, about 25% by chance). Figure 11 shows that estimates are compressed towards 15–45 months: early progressions are overestimated and late ones underestimated. Agreement with the recorded window is 12% in group A, 60% in B, 37% in C, and 29% in D. Several estimators were compared on the same progressors, so the chosen row is mildly optimistic.

Table 6: Estimating the month of progression on held-out progressors.

Estimator	MAE	Median AE	≤ 12 mo	Window	IQR cov.
Constant (training progressor median)	22.9	16.0	37%	19%	49%
Conditional median, $\tau = 60$	23.2	18.2	33%	18%	37%
Conditional median, $\tau = 120$	27.9	24.6	23%	39%	41%
Conditional median, $\tau = 180$	59.2	62.0	7%	39%	33%
Log-normal AFT on progressors	20.6	12.9	48%	34%	55%
Log-logistic AFT on progressors	**20.1**	**12.9**	**48%**	**33%**	**57%**

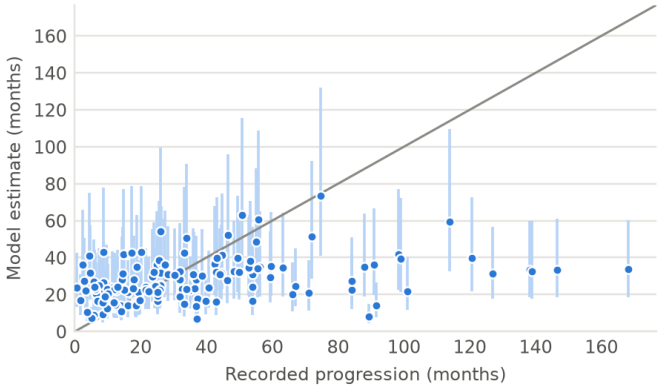


Figure 11: Model-estimated against recorded progression month. Bars are the model's interquartile ranges, and the diagonal is perfect agreement.

4.7.5 Neuro-Symbolic Paths

The explain stage enumerated 59,598 candidate paths for 1,033 patients in about 9 minutes. The other 50 patients have no path, because none of their altered genes has a PrimeKG pathway, drug, or breast-disease link. The prominence cutoff is an NS score of 0.523 (Figure 12). 27.5% of all candidate paths, and 35% of the paths shown in reports, ground at least one learned formula. Most patients (801) receive $K = 3$ paths, because they have fewer than three prominent paths and are topped up; 62 patients have five prominent paths. A median of three weighted formulas fires per patient, and 9.7% of patients have none. Figure 13 shows the four prominent paths of patient TCGA-BH-A1EV (group A, progression at 12.0 months). All four ground formulas that raise the hazard. The strongest runs from the amplified gene NFASC to the pathway *Neurofascin interactions* ($\sum w = +0.134$ over two formulas).

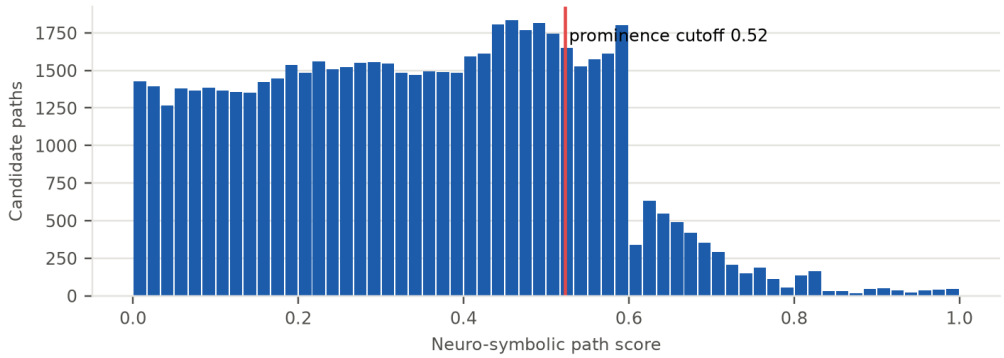


Figure 12: Distribution of neuro-symbolic path scores, with the prominence cutoff at the cohort's 80th percentile.

4.7.6 Patient Reports

The system wrote 1,083 patient reports, five group summaries, four combined group PDFs, and a cohort overview. Figures 14 to 16 show the full report of TCGA-BH-A1EV. Figure 17 shows the first page of a patient without recorded progression, whose outcome tile states the follow-up length and that censoring is not proof of remission. Figure 18 shows the summary page of group A.

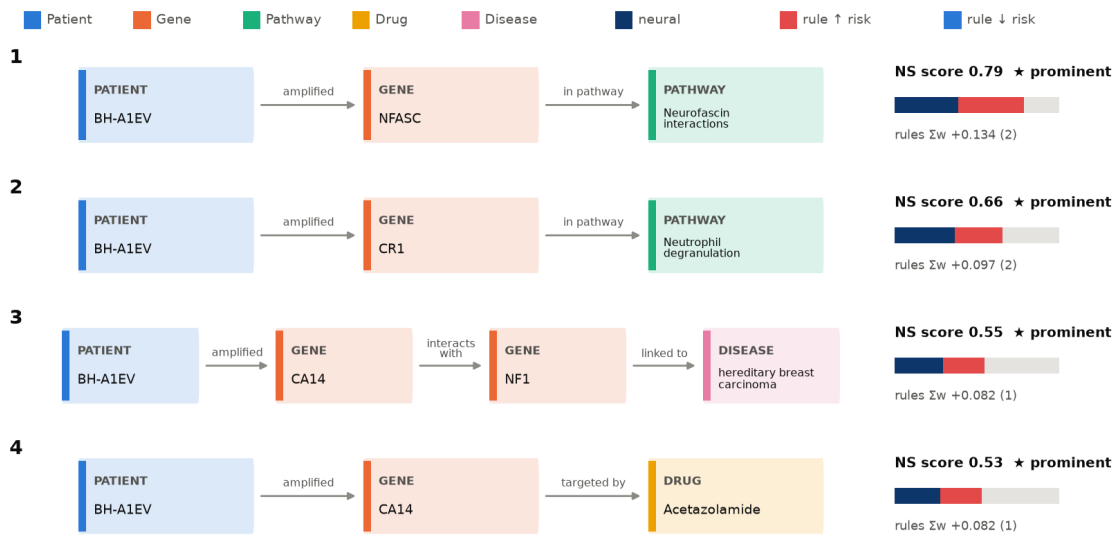


Figure 13: Top-K paths of patient TCGA-BH-A1EV. The bar on the right splits each NS score into its neural part (navy) and its rule part (red: the grounded formulas raise the hazard; blue: they lower it).

4.8 Discussion

Across five milestones, every attempt to raise discrimination with graph structure ended at the same ceiling: relation-aware layers, concordance objectives, structure learning, and survival training on the full follow-up. With 145 events, nodal status, stage, age, and subtype carry most of the prognostic signal, and somatic alterations on a 500-gene panel add little beyond it. The PrimeKG facts are not uninformative. Several pathway facts separate carriers with hazard ratios around 1.5 on the training folds, and the neural and logic risks together reach a C-index of 0.642 without any clinical input. Their information, however, overlaps with the clinical variables. The logic reasoners show the same pattern from the symbolic side. Moving from crisp IF–THEN rules to weighted, first-order, knowledge-graph-grounded formulas changes *what* the system can say about a patient much more than *how well* it ranks patients.

The neuro-symbolic system therefore earns its place through explanation. Each report states which PrimeKG facts the graph model attended to, which weighted

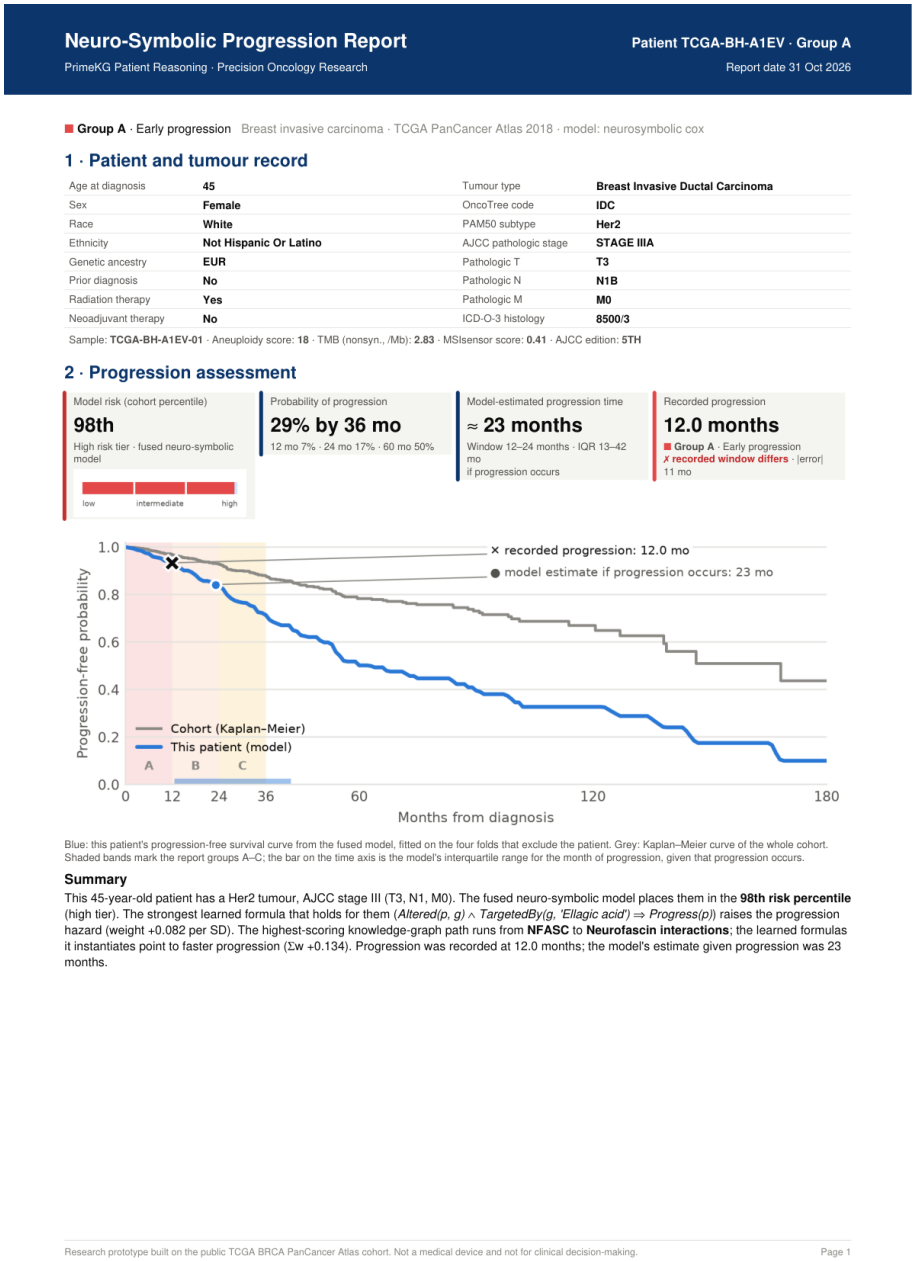
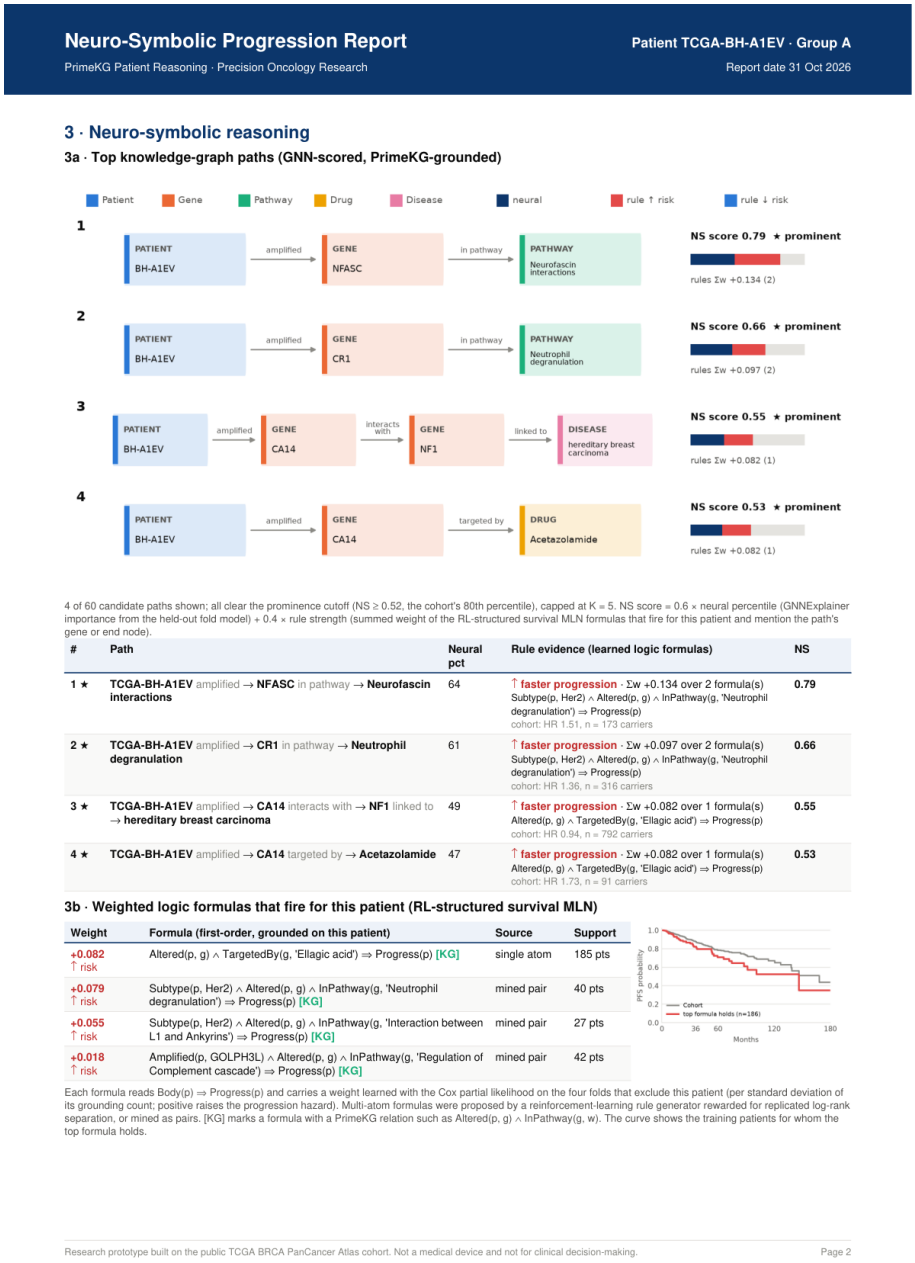


Figure 14: Patient report, page 1: record, progression assessment, survival curve, and summary.



Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1EV · Group A

Report date 31 Oct 2026

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.52, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × rule strength (summed weight of the RL-structured survival MLN formulas that fire for this patient and mention the path's gene or end node).

#	Path	Neural pct	Rule evidence (learned logic formulas)	NS
1 ★	**TCGA-BH-A1EV** amplified → **NFASC** in pathway → **Neurofascin interactions**	64	↑ faster progression · Σw +0.134 over 2 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) cohort: HR 1.51, n = 173 carriers	0.79
2 ★	**TCGA-BH-A1EV** amplified → **CR1** in pathway → **Neutrophil degranulation**	61	↑ faster progression · Σw +0.097 over 2 formula(s) Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) cohort: HR 1.36, n = 316 carriers	0.66
3 ★	**TCGA-BH-A1EV** amplified → **CA14** interacts with → **NF1** linked to → **hereditary breast carcinoma**	49	↑ faster progression · Σw +0.082 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) cohort: HR 0.94, n = 792 carriers	0.55
4 ★	**TCGA-BH-A1EV** amplified → **CA14** targeted by → **Acetazolamide**	47	↑ faster progression · Σw +0.082 over 1 formula(s) Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) cohort: HR 1.73, n = 91 carriers	0.53

3b · Weighted logic formulas that fire for this patient (RL-structured survival MLN)

Weight	Formula (first-order, grounded on this patient)	Source	Support
+0.082 ↑ risk	Altered(p, g) ∧ TargetedBy(g, 'Ellagic acid') ⇒ Progress(p) [KG]	single atom	185 pts
+0.079 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Neutrophil degranulation') ⇒ Progress(p) [KG]	mined pair	40 pts
+0.055 ↑ risk	Subtype(p, Her2) ∧ Altered(p, g) ∧ InPathway(g, 'Interaction between L1 and Ankyrins') ⇒ Progress(p) [KG]	mined pair	27 pts
+0.018 ↑ risk	Amplified(p, GOLPH3L) ∧ Altered(p, g) ∧ InPathway(g, 'Regulation of Complement cascade') ⇒ Progress(p) [KG]	mined pair	42 pts

Each formula reads Body(p) ⇒ Progress(p) and carries a weight learned with the Cox partial likelihood on the four folds that exclude this patient (per standard deviation of its grounding count; positive raises the progression hazard). Multi-atom formulas were proposed by a reinforcement-learning rule generator rewarded for replicated log-rank separation, or mined as pairs. [KG] marks a formula with a PrimeKG relation such as Altered(p, g) ∧ InPathway(g, w). The curve shows the training patients for whom the top formula holds.

Figure 15: Patient report, page 2: top-K knowledge-graph paths and the weighted logic formulas that fire.

Neuro-Symbolic Progression Report

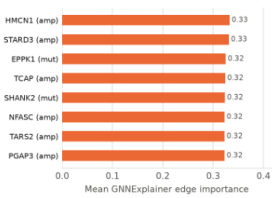
PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1EV · Group A

Report date 31 Oct 2026

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
BRCA2 · driver	High-level amplification	GISTIC ±2	14	0.31
HMCN1	High-level amplification	GISTIC ±2	0	0.33
STARD3	High-level amplification	GISTIC ±2	1	0.33
EPPK1	Mutation	p.L1474=	0	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
SHANK2	Mutation	p.P1240T	1	0.32
NFASC	High-level amplification	GISTIC ±2	3	0.32
TARS2	High-level amplification	GISTIC ±2	2	0.32
PGAP3	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32



46 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (−2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

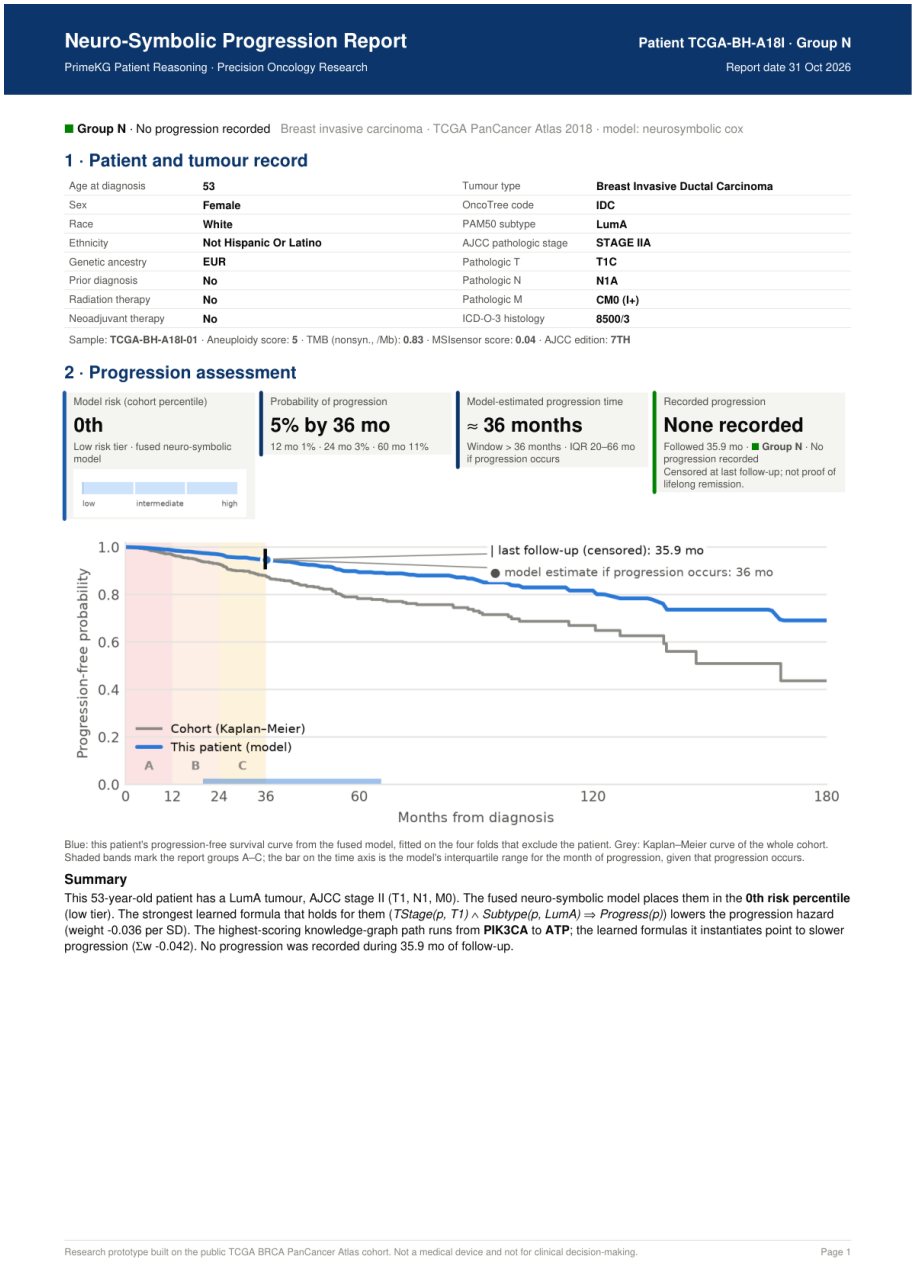
Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A1HL	38	III	LumB	0.70	no progression, followed 10.4 mo
TCGA-A8-A07I	69	III	Her2	0.66	no progression, followed 14.0 mo
TCGA-B6-A0IQ	40	III	Basal	0.62	no progression, followed 140.9 mo
TCGA-B6-A0WX	40	III	Basal	0.62	progressed at 18.5 mo (group B)
TCGA-A8-A09B	58	III	LumA	0.62	no progression, followed 12.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the logic-engine risk (RL-structured survival MLN), all fitted on the four folds that exclude this patient. In cross-validation the GNN and logic terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The paths show which PrimeKG facts the graph model attended to, and the weighted logic formulas state which facts, alone or combined, shift the progression hazard in the training folds; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 48% of held-out progressors and in the recorded window for 33%; the recorded month is always shown beside it.

Figure 16: Patient report, page 3: molecular profile, PrimeKG drug links, similar patients, and reading notes.



Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A18I · Group N

Report date 31 Oct 2026

Group N · No progression recorded Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**53**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**CM0 (I+)**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A18I-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **0.83** · MSIsensor score: **0.04** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)
0th
Low risk tier · fused neuro-symbolic model
low · intermediate · high

Probability of progression
5% by 36 mo
12 mo 1% · 24 mo 3% · 60 mo 11%

Model-estimated progression time
≈ 36 months
Window > 36 months · IQR 20–66 mo if progression occurs

Recorded progression
None recorded
Followed 35.9 mo · **Group N** · No progression recorded
Censored at last follow-up; not proof of lifelong remission.

Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 53-year-old patient has a LumA tumour, AJCC stage II (T1, N1, M0). The fused neuro-symbolic model places them in the **0th risk percentile** (low tier). The strongest learned formula that holds for them ($TStage(p, T1) \wedge Subtype(p, LumA) \Rightarrow Progress(p)$) lowers the progression hazard (weight -0.036 per SD). The highest-scoring knowledge-graph path runs from **PIK3CA** to **ATP**; the learned formulas it instantiates point to slower progression (Σw -0.042). No progression was recorded during 35.9 mo of follow-up.

Research prototype built on the public TCGA BRCA PanCancer Atlas cohort. Not a medical device and not for clinical decision-making. Page 1

Figure 17: First page of a report in group N (no progression recorded).

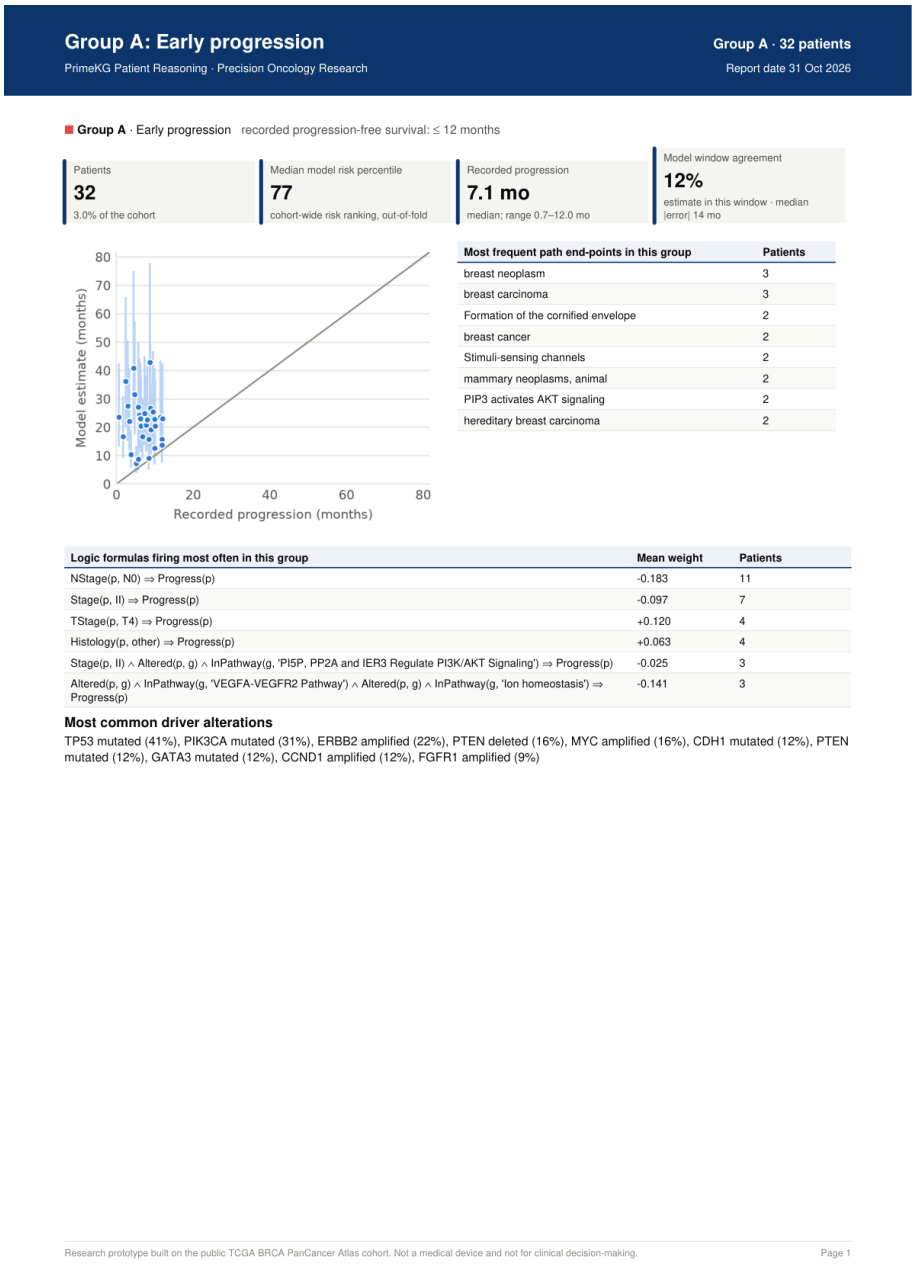


Figure 18: Summary page of group A, progression within 12 months.

logic formulas ground those paths, and which formulas fire for the patient. It does so without claiming accuracy that the evaluation does not support.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

We built a complete pipeline from raw TCGA tables to patient-specific knowledge graphs fused with PrimeKG, benchmarked seven graph neural network architectures under three objectives, and turned the result into a neuro-symbolic progression model with per-patient reports. Its symbolic engine replaces IF–THEN rule induction with a survival Markov logic network whose first-order, knowledge-graph-grounded formulas are proposed by reinforcement learning and weighted by the Cox partial likelihood. We compared it against probabilistic soft logic with collective inference, RL rules alone, and RuleKit. The fused model reaches a concordance index of 0.664, statistically tied with the best clinical model, and it estimates the month of progression to within a year for about half of progressors. Every prediction is out-of-fold and leakage-checked. Every one of the 1,083 reports explains its estimate with explicit knowledge-graph paths and weighted logic formulas, and places the patient in a progression-timeline group.

The central finding is that, on this cohort, knowledge-graph structure adds explanation rather than accuracy. We report this result as it is, and the system is designed around it.

5.2 Limitations

- Discrimination saturates at a C-index of about 0.67 with 145 events. The neuro-symbolic terms carry close to zero weight once clinical variables are present.
- The month-of-progression estimate is coarse. Late progressions (beyond five years) are hard to place.
- The logic engine adds reasoning, not accuracy: Markov logic matches RuleKit within noise, PSL's collective rule found no signal, and RL rules alone overfit their search.
- Paths explain the GNN, which is the weaker component. They are evidence, not causal claims.
- PrimeKG drug links include enzymes, carriers, and transporters, and are not treatment recommendations.
- One cohort, one seed, and no external validation of the reports. The fusion uses cross-validated stacking.

5.3 Future Work

- **More modalities**: add expression, methylation, and protein data as node features, which the current graph excludes by design.
- **Pretrained knowledge-graph embeddings**: initialise PrimeKG node embeddings from TransE or a pretrained biomedical model instead of training them from scratch.
- **Richer RL reasoning**: let the policy walk multi-hop PrimeKG relation paths (as in MINERVA) rather than choose atoms, and train it jointly with the MLN weights (as in RNNLogic).

- **Joint MLN inference**: add formulas that link patients (shared pathway disruption) and run full MLN inference instead of the factorised per-patient score.
- **External validation**: apply the model and reports to METABRIC or GSE2990, whose relapse-free survival times are already extracted.
- **Clinician evaluation**: have oncologists rate the usefulness and correctness of the reports.

Appendix A

Reproducibility

All code is in the project repository. From the repository root, the commands below build the Milestone 1 graphs, run the Milestone 2 benchmark, run the five neuro-symbolic stages (neural, symbolic, logic, fusion, explain), write the reports and the notebook, draw the figures of this report, and run the 58 unit tests:

```
pip install -r requirements.txt
python scripts/build_patient_graphs.py
python scripts/train_and_evaluate_gnns.py \
    --endpoint pfs_36m --skip-sensitivity
python scripts/run_neurosymbolic_pipeline.py
python scripts/build_patient_reports.py --workers 4
python scripts/build_neurosymbolic_notebook.py
python report/make_figures.py
python -m unittest discover -s tests
```

The neuro-symbolic pipeline caches each stage under `outputs/neurosymbolic/`. The reports are written to `outputs/patient_reports/`, in one folder per group. The method is documented in `docs/neurosymbolic_reports.md`, and the executed walkthrough is `notebooks/neurosymbolic_patient_reports.ipynb`. RuleKit needs a Java runtime, version 11 or later.

Table 7: Main source modules of the neuro-symbolic system.

Module	Responsibility
`src/dataset_fusion.py`	PrimeKG slice, neighbourhood expansion, fused PyG graphs
`src/gnn_models.py`	seven GNN architectures and the clinical residual
`src/survival.py`	comparable pairs, C-index, Cox and C-index objectives
`src/neurosymbolic/features.py`	clinical, genomic, and PrimeKG predicates
`src/neurosymbolic/rules.py`	RuleKit baseline and readable rule text
`src/neurosymbolic/logic/atoms.py`	first-order ground atoms, soft survival targets
`src/neurosymbolic/logic/mln.py`	survival Markov logic network (Cox-MLN)
`src/neurosymbolic/logic/psl.py`	survival probabilistic soft logic
`src/neurosymbolic/logic/rl.py`	policy-gradient rule generator
`src/neurosymbolic/logic/runner.py`	reasoner comparison and engine choice
`src/neurosymbolic/kg_evidence.py`	vectorised log-rank evidence for PrimeKG facts
`src/neurosymbolic/paths.py`	path scoring and top-K selection
`src/neurosymbolic/timing.py`	report groups and the progression-month estimators
`src/neurosymbolic/report_pdf.py`	patient reports, group summaries, cohort overview

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